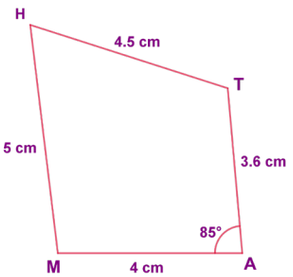
5.10.3 Construct a quadrilateral when its 4 sides and one angle are given

Example 5.23

Construct a quadrilateral $MATH$ with MA=4 cm, AT= 3.6 cm, TH = 4.5 cm, MH= 5 cm and $\angle A = 85°$. Also find its area.

Solution:

Given:
MA=4 cm, AT= 3.6 cm,
TH = 4.5 cm, MH= 5 cm and $\angle A = 85°$



Rough diagram

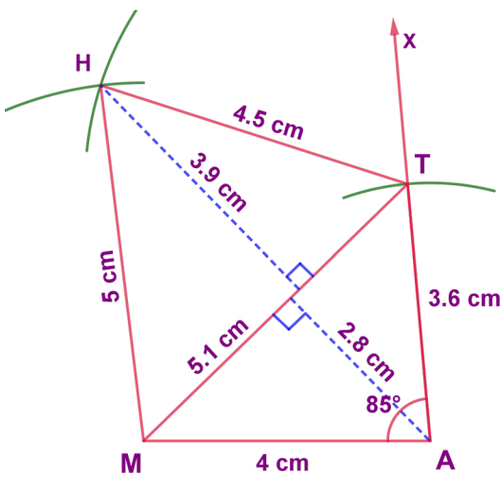


Fig. 5.42

Steps:

1. Draw a line segment MA = 4 cm.
2. Make $\angle A = 85°$.
3. With A as centre, draw an arc of radius 3.6 cm. Let it cut the ray AX at T.
4. With M and T as centres, draw arcs of radii 5 cm and 4.5 cm respectively and let them cut at H.
5. Join MH and TH.
6. $MATH$ is the required quadrilateral.

Calculation of Area:

Area of the quadrilateral $MATH$ $= \frac{1}{2} \times d \times (h_1 + h_2)$ *sq. units*

$= \frac{1}{2} \times 5.1 \times (3.9 + 2.8)$

$= 2.55 \times 6.7 = 17.09\ cm^2$

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